Lab 4 – Vibe Coding Chat History

Repository: SETAPESU26/07_catch_falling

Student ID: PES1UG23AM245

User

help me do it

Assistant

The Lab 4 handout requires cloning/forking the assigned repository, running the Python game, recording a 10-second video before changes, fixing the broken code, adding the features listed in the README one by one, recording an after video, pushing the code, and keeping separate commits for each task. The assigned repository is SETAPESU26/07_catch_falling and the student ID is PES1UG23AM245.

User

the game is giving ModuleNotFoundError: No module named 'pygame'

Assistant

We identified that Pygame was missing. The active Python was C:\msys64\ucrt64\bin\python.exe and it did not have pip. We checked installed Python versions and found a working Python 3.11 installation at C:\Users\samhi\AppData\Local\Programs\Python\Python311\python.exe. Pygame was installed using that Python, and the game started successfully.

User

the game started what do i do ?

Assistant

The next step was to record the required 10-second BEFORE video before changing any code. The recording was intended to demonstrate the original broken behavior.

User

before video done

Assistant

Task 1 was started. The two Task 1 bugs were identified: collision detection only checked horizontal position, and objects were removed from the same list while iterating over it. The requested fix was to require both horizontal and vertical overlap for a catch and to safely collect caught objects before removing them.

User

u r a vibe coding tool right help me do task 1

Assistant

The assistant acted as the Vibe Coding tool and provided the Task 1 implementation. In game/collision.py, collision detection was changed to check horizontal and vertical overlap. In game/game_engine.py, caught objects were collected first and removed after iteration to avoid list mutation while iterating.

User

shows git diff for Task 1

Assistant

The Task 1 diff was reviewed and confirmed to contain the intended changes in game/collision.py and game/game_engine.py. The user was asked to run git diff --check, test the game, and then commit Task 1 separately.

User

done i recorded the game again now what

Assistant

Task 1 was considered complete after testing and recording. The user was instructed to commit and push Task 1 separately with git add game/collision.py game/game_engine.py, git commit -m "Fix falling object collision detection", and git push origin main.

User

i added the prompt and made the necessary changes as well now what?

Assistant

Task 2 was tested next. The goal was smooth continuous basket movement and keeping the entire basket inside the screen. The user was instructed to test the left and right boundaries, run git diff --check, and commit Task 2 separately.

User

shows Task 2 git diff

Assistant

The Task 2 diff was reviewed. It used a movement variable for continuous left/right movement and calculated half the basket width when clamping the basket position. This was confirmed as the correct approach for keeping the whole basket inside the screen.

User

task 3

Assistant

Task 3 was started. The Vibe Coding prompt requested controlled object spawning: varied horizontal positions, varied spawn intervals, positions kept inside the playable area, avoidance of repeated horizontal positions, and a maximum active-object limit.

User

next

Assistant

Task 4 was started. The requested feature was a temporary basket speed boost activated during gameplay, with increased speed, a visible indication, automatic expiration, restoration of normal speed, and preservation of existing boundaries and game behavior.

User

shows Task 4 git diff

Assistant

The diff was inspected. The basket had normal and boosted speed values, but the remaining game_engine.py and main.py changes were checked before committing to verify activation, timer/expiration, and the visual indicator.

User

shows git status

Assistant

git status showed modifications in game/basket.py, game/falling_object.py, game/game_engine.py, and main.py. The assistant asked for the complete diff to verify that Task 4 did not introduce unrelated changes and that Tasks 1–3 remained intact.

User

shows full game_engine.py and main.py diff

Assistant

The diff showed Task 3 spawning constants and logic, including a maximum spawn interval, minimum spawn distance, maximum active objects, and boost duration. main.py was updated to mention Space as the temporary speed boost control. The assistant requested inspection of the remaining boost logic before final commit.

User

next

Assistant

The user was instructed to run the completed game and test the speed boost, Task 3 spawning behavior, basket boundaries, and the expiration of the boost. After successful testing, the final GitHub commits and final after-video would be prepared.

User

lab4 submission folder

Assistant

The recommended Lab-4 submission structure was: Lab-4/before.mp4, Lab-4/after.mp4, Lab-4/chat_history.pdf, and Lab-4/updated_code/ containing main.py, requirements.txt, and the game source files. The final after video should show the completed game after Tasks 1–4.

End of chat history.